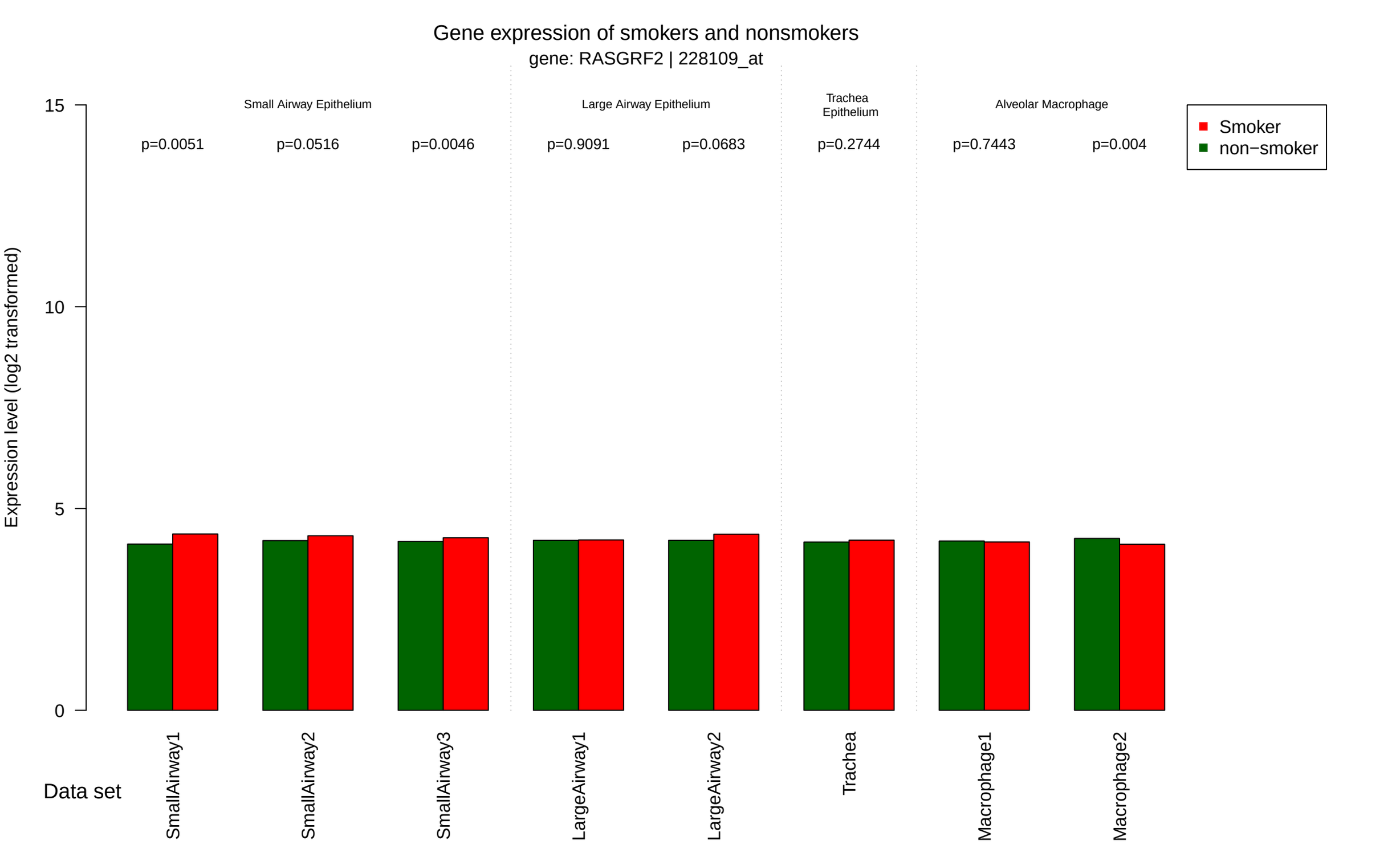
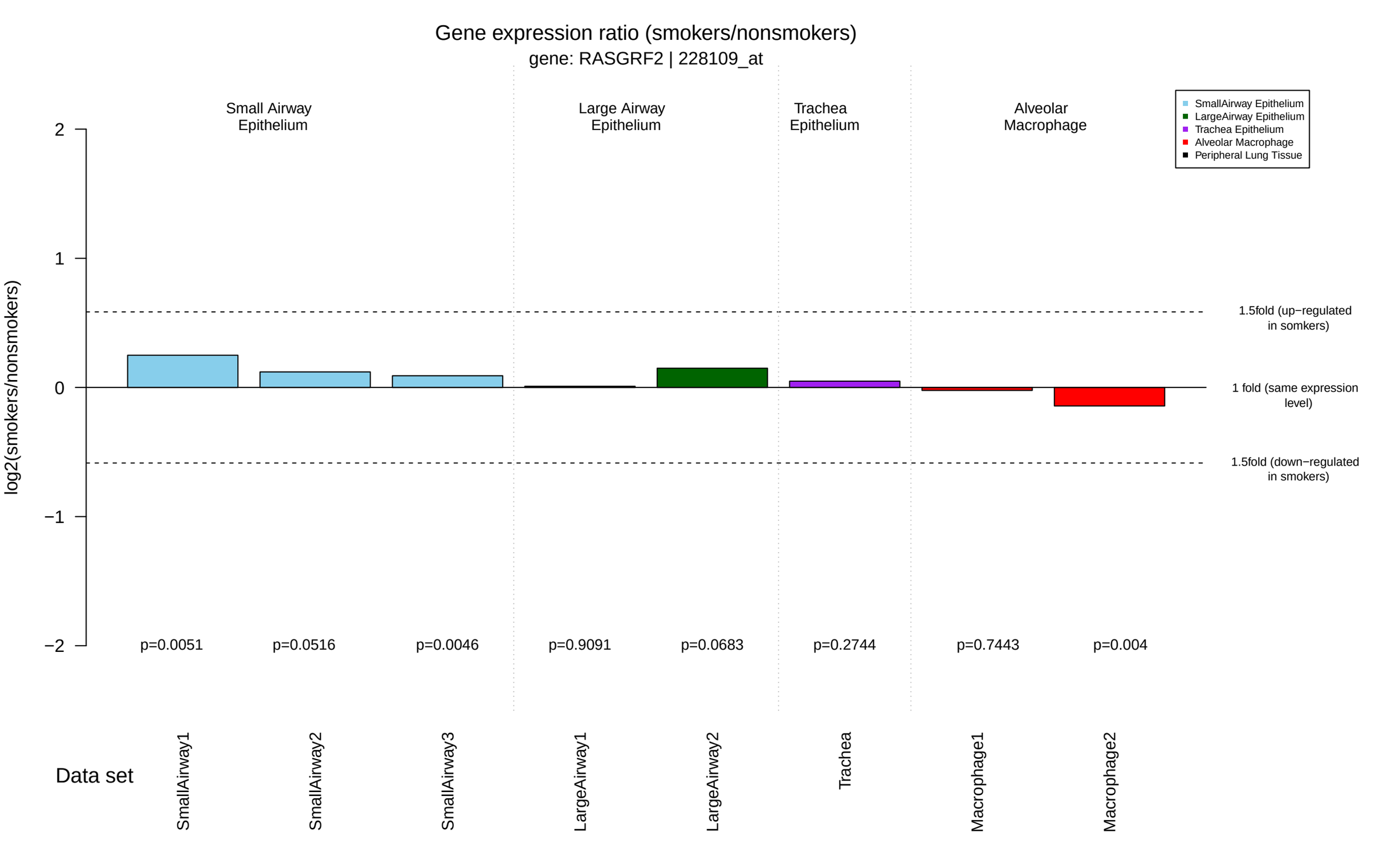
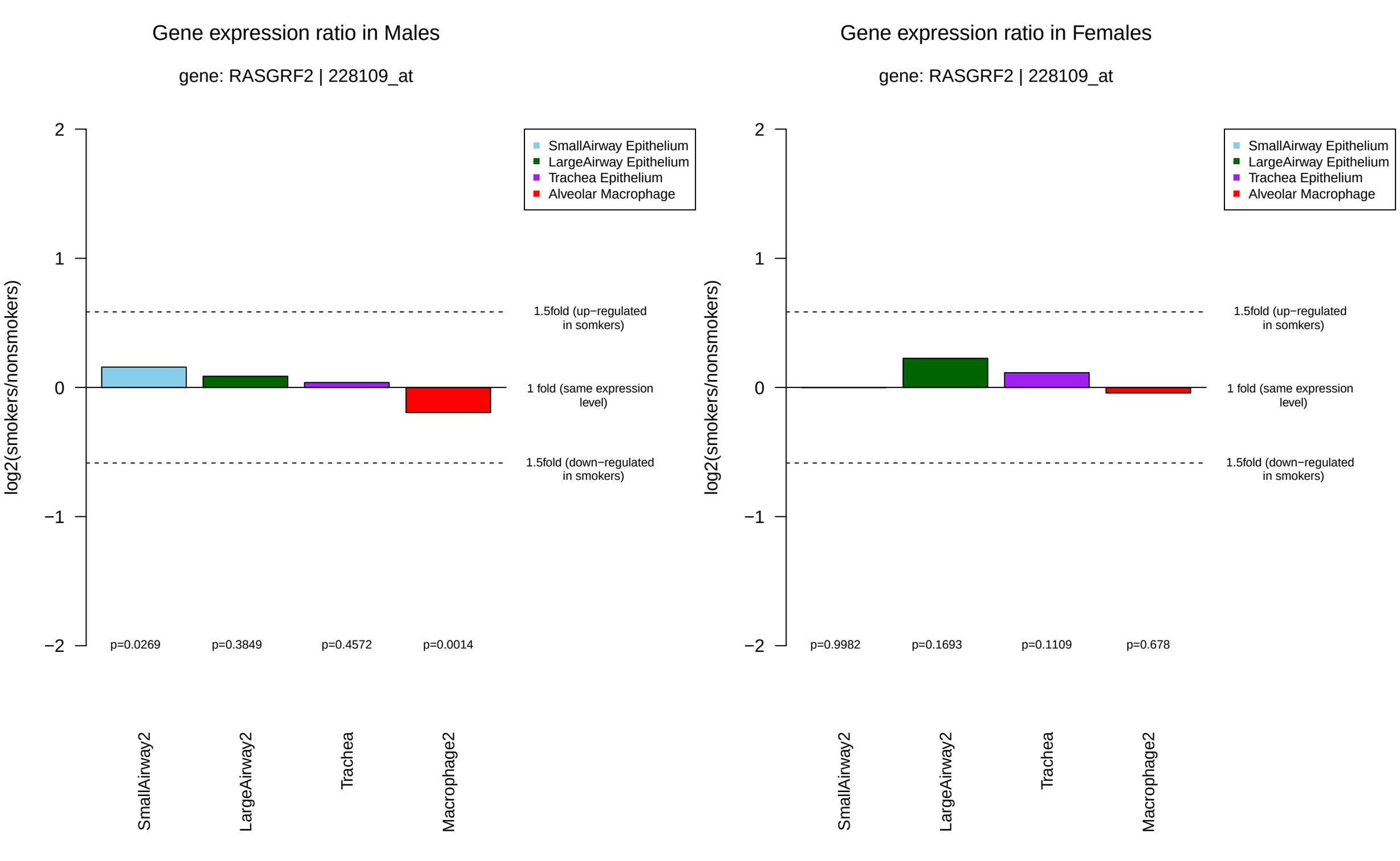
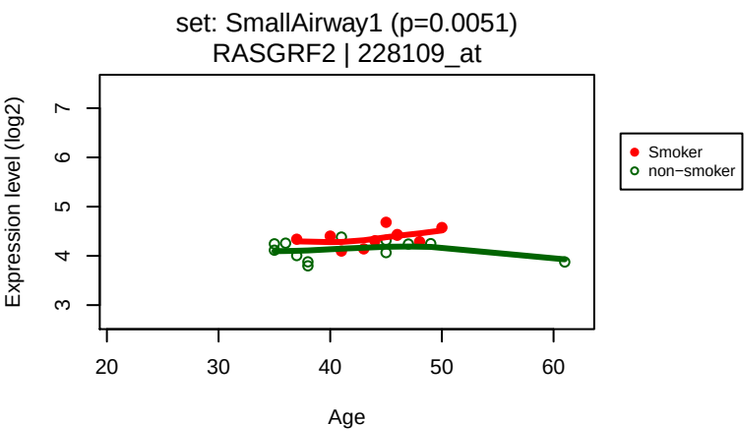
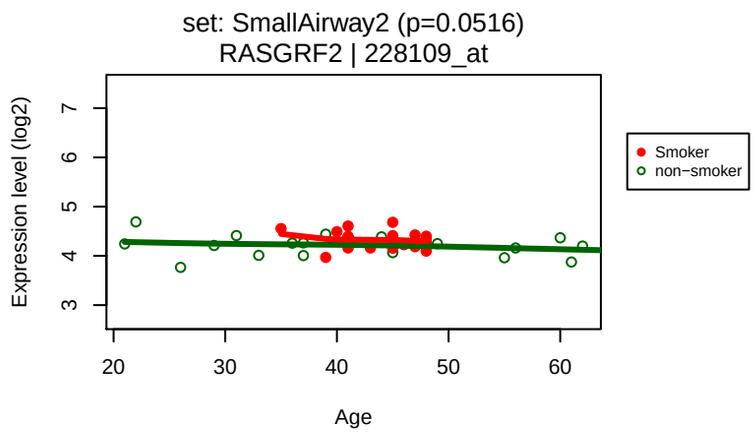
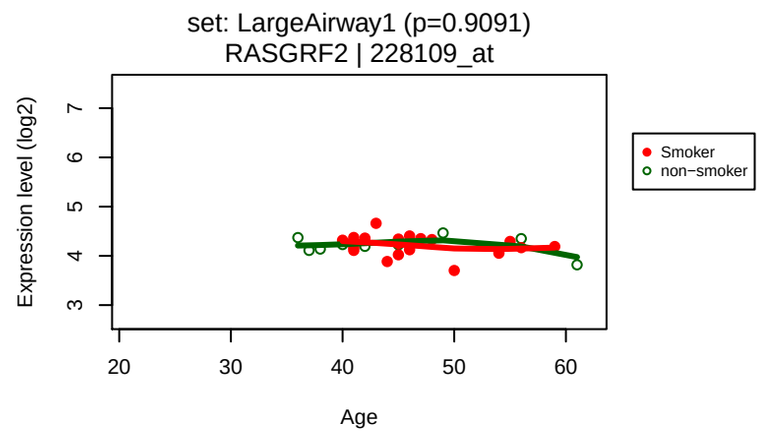

Gene expression of smokers and nonsmokers
gene: RASGRF2 | 228109_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0051
p=0.0516
p=0.0046
p=0.9091
p=0.0683
p=0.2744
p=0.7443
p=0.004
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

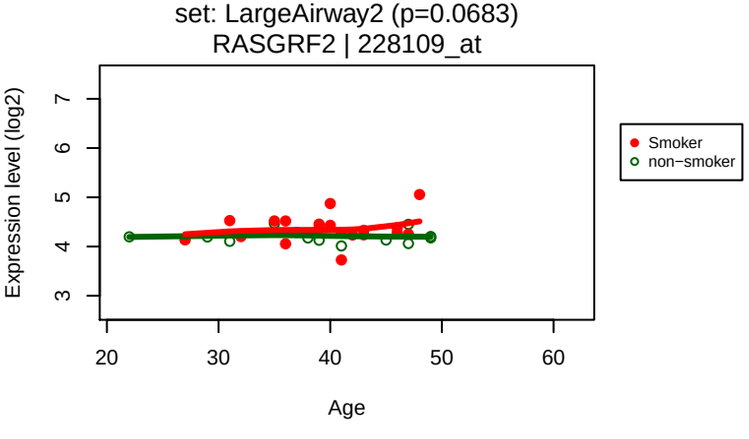

Gene expression ratio (smokers/nonsmokers)
gene: RASGRF2 | 228109_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0051
p=0.0516
p=0.0046
p=0.9091
p=0.0683
p=0.2744
p=0.7443
p=0.004
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

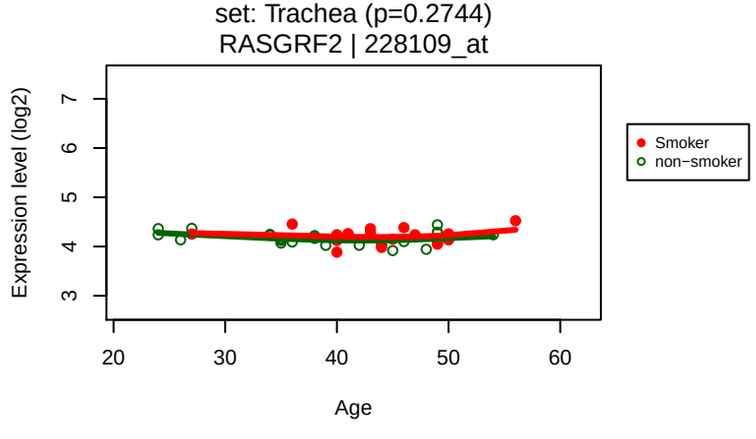

Gene expression ratio in Males
gene: RASGRF2 | 228109_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0269
p=0.3849
p=0.4572
p=0.0014
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RASGRF2 | 228109_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9982
p=0.1693
p=0.1109
p=0.678
SmallAirway2
LargeAirway2
Trachea
Macrophage2

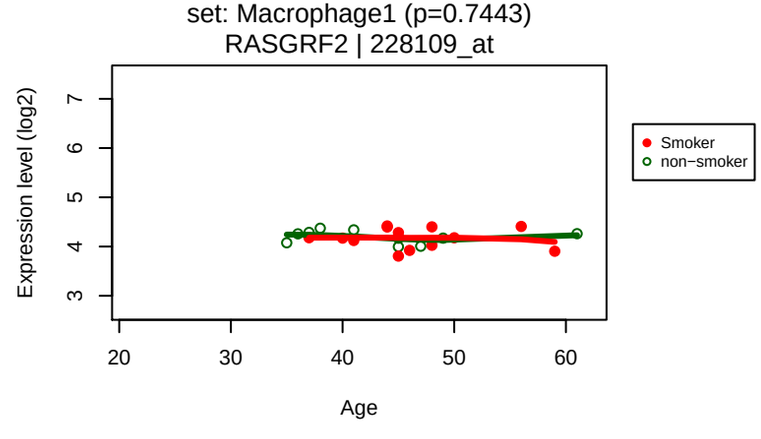

set: SmallAirway1 (p=0.0051)
RASGRF2 | 228109_at
Expression level (log2)
Age
Smoker
non−smoker

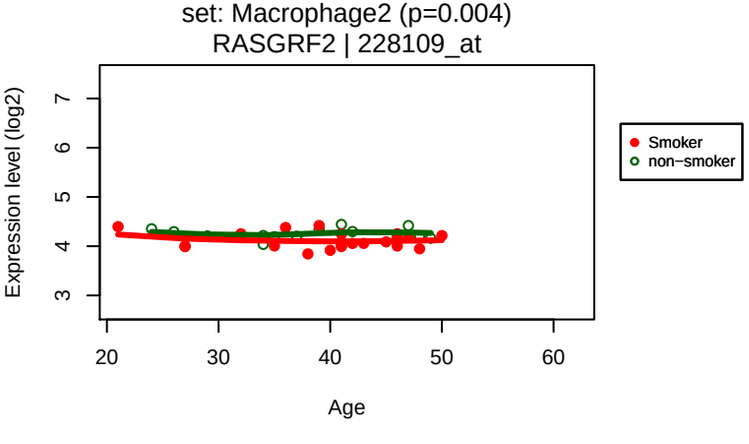

set: SmallAirway2 (p=0.0516)
RASGRF2 | 228109_at
Expression level (log2)
Age
Smoker
non−smoker

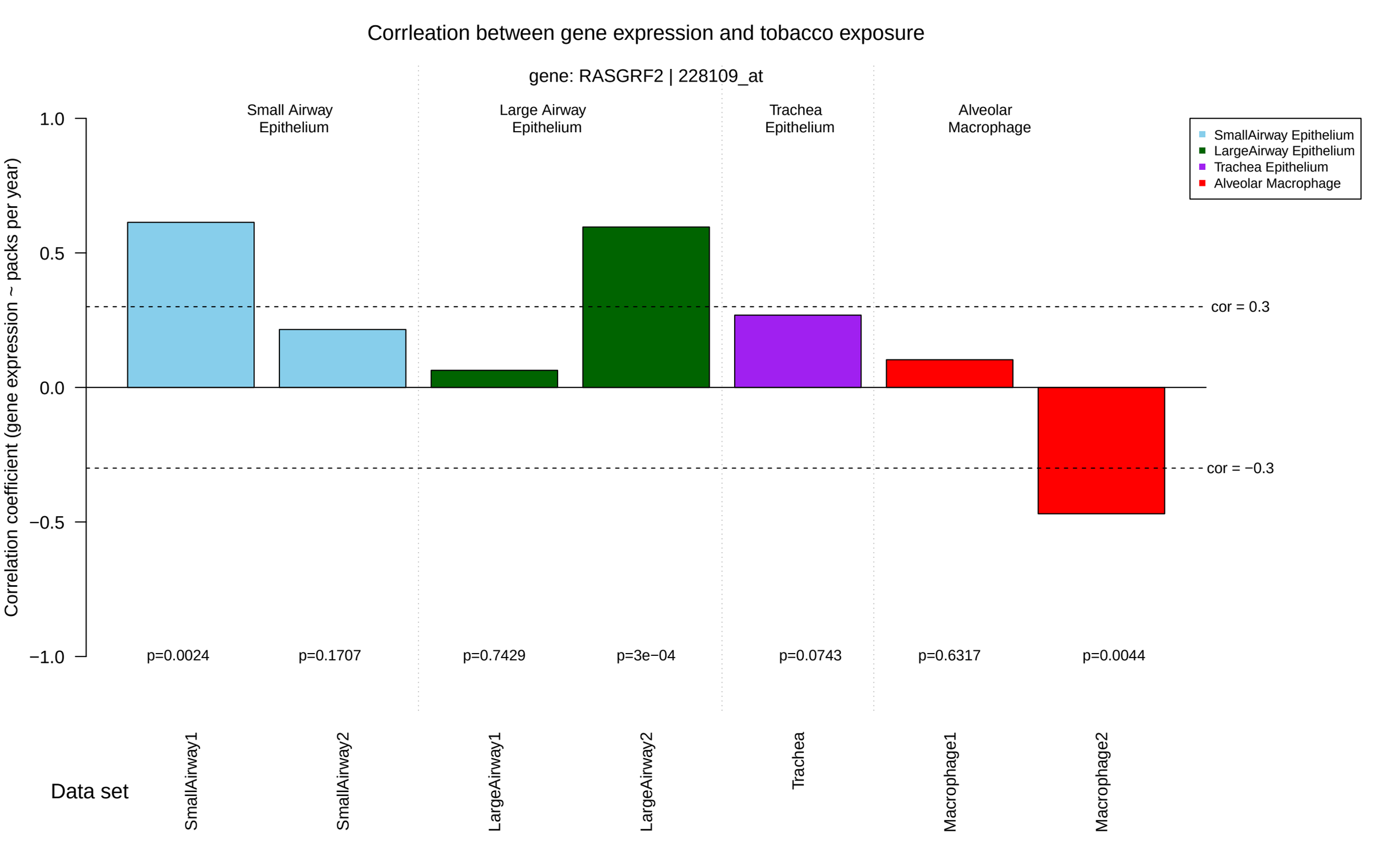

set: LargeAirway1 (p=0.9091)
RASGRF2 | 228109_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0683)
RASGRF2 | 228109_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2744)
RASGRF2 | 228109_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7443)
RASGRF2 | 228109_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.004)
RASGRF2 | 228109_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RASGRF2 | 228109_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0024
p=0.1707
p=0.7429
p=3e−04
p=0.0743
p=0.6317
p=0.0044
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2